

AUTOSAR Body Control Module (BCM) High-Level Design (Version 1.0)

[SAMPLE / TEST DATA FOR DEMONSTRATION ONLY] | Document Version: 1.0

1. System Purpose & Functional Scope

The Body Control Module (BCM) manages vehicle interior lighting, door locks, and window position control. It integrates Application SWCs with the BSW CAN communication stack.

2. Application Components

Components defined include **LightingControl_SWC** and **DoorLock_SWC**. **LightingControl_SWC** receives headlight switch status over CAN and drives PWM headlight relays.

3. Interface & Port Configuration

DoorLock_SWC exposes **RPort_DoorStatus** to read latch microswitch state and **PPort_LockActuator** to trigger solenoids. Interface **If_DoorState** conveys lock state via signal **Sig_DoorLock_status**. Legacy interface **If_LegacyState** conveys legacy diagnostic signals via **Sig_Legacy_status**.

4. Basic Software Stack

The BSW stack includes **EcuM**, **BswM**, **CanIf**, and **Com**.